

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)	Electrons move at right angles to B -field OR LHR mentioned OR accept Bqv or BIl force (anything to do with magnetic force acting on electrons) (1) Force towards back face or shown clearly in diagram (1) (-ve) charge build (up on back face) (1) Electric field hence voltage (ignore “inducing” = neutral) OR accept equilibrium when mag force = electric force OR accept charge build-up <u>clearly</u> linked to Hall voltage (1)		4		4		4
		(ii)	$Ee = Bqv$ or similar seen e.g. $Eq = BIl$ or conservation of energy stated (1) $E = \frac{V}{d}$ substituted OR $E = \frac{V}{d}$ and $q = It$ and $v = \frac{l}{t}$ substituted or $Vq = Bqvd$ i.e. conservation of energy applied (1) Algebra seen to give correct answer (1)	1	1 1		3	2	
		(iii)	Any 3 numbers substituted to obtain the 4 th (1) Correct answer 4 th number & conclusion seen (1) ($7.26 \times 10^{-5} \text{ m s}^{-1}$ is correct drift velocity) 5.4 mm is calculated d , 65.5 nV is calculated Hall pd, 0.187 T is calculated B -field)			2	2	2	2
		(iv)	Rearrangement of $I = nAve$ ($n = \frac{I}{Ave}$) OR $V_H = \frac{BI}{net}$ known (1) Correct area method i.e. $5.2 \times 0.4 (\times 10^{-6})$ OR $n = \frac{BI}{V_H et}$ (1) Correct answer ($3.28 \times 10^{28} \text{ m}^{-3}$) OR 3.4×10^{28} if they use 0.07 mm s^{-1} (beware ecf may be available for a candidate that uses a previously penalised value instead of the given values)(1)		3		3	3	3